AB CLOUD QUANTUM RESEARCH · MONOGRAPH SERIES

VOLUME TS2

TS2. The volatility theorem

Supplementary study 'the machine with no long-term memory': three spinor phase models θ = π/2 · π/3 · π/7 — volatile state only, real-time steps, zero storage

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Исаев Исхак Хамзатович · version 1.0.0 · 2026-10-11

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AB Cloud Quantum Research · supplementary study · том TS2 · версия 1.0.0 · 2026-10-11

Contents

TS2. The volatility theorem **2**

Certificate ES2 (state volatility) **2**

Boundaries and honest limitations **3**

TS2. The volatility theorem

Theorem TS2 (ES2).

Let the machine have a single register — a phase $s \in G_\theta$ — and reset: $G_\theta \to \{0\}$ erase it to the canonical state. Then for any prior P on G_θ: (i) the mutual information $I(S_{pre}; S_{post}) = 0$ — the post-state is degenerate; (ii) no observer function $g(S_{post})$ recovers S_{pre} with probability above $\max_i P(S_{pre} = i)$, and for uniform P — above 1/N (measured: 0.2501 vs 1/N = 0.2500 for N = 4); (iii) about time t the state carries exactly $s_t \equiv (s_0 + t) \pmod N$: the share of consistent $t' \in [0, T)$ is 1/N (measured 0.250000 at $T = 10^6$). Corollary: such a machine has no long-term memory by construction — the state carries $\log_2 N$ bits regardless of the runtime.

> ***Proof.*** *(i) S_{post} is constant: $H(S_{post}) = 0$, and $I(X; Y) = 0$ whenever Y is degenerate. (ii) For any strategy g: $P[g(S_{post}) = S_{pre}] = \Sigma_i P(i) \cdot 1\{g(0) = i\} \le \max_i P(i)$; for uniform P this is 1/N — the classical blind-guessing bound. (iii) From the deterministic dynamics $s_t \equiv (s_0 + t) \pmod N$ the set of consistent times is a residue class mod N with asymptotic share 1/N in [0, T).* ∎

Certificate ES2 (state volatility)

Campaign ES2 ran with pre-registered thresholds; verdict: PASS.

All checks were fixed before the run; raw data (JSON + CSV) is committed under data/ and reproduced by the experiments/ scripts.

Table 1. Certificate checks (campaign).

Check	Result	Threshold
C1 reset irreversibility	0.2501 / 0.1674 / 0.0717	≤ 1/N + 3σ (erratum ES2a)
C2 I(пре; пост)	0 exactly (degenerate post)	= 0
C3 no-reset control	deterministic, 1.0	= 1.0
C4 время → mod N	share of t′ = 1/N (±1%)	≈ 1/N

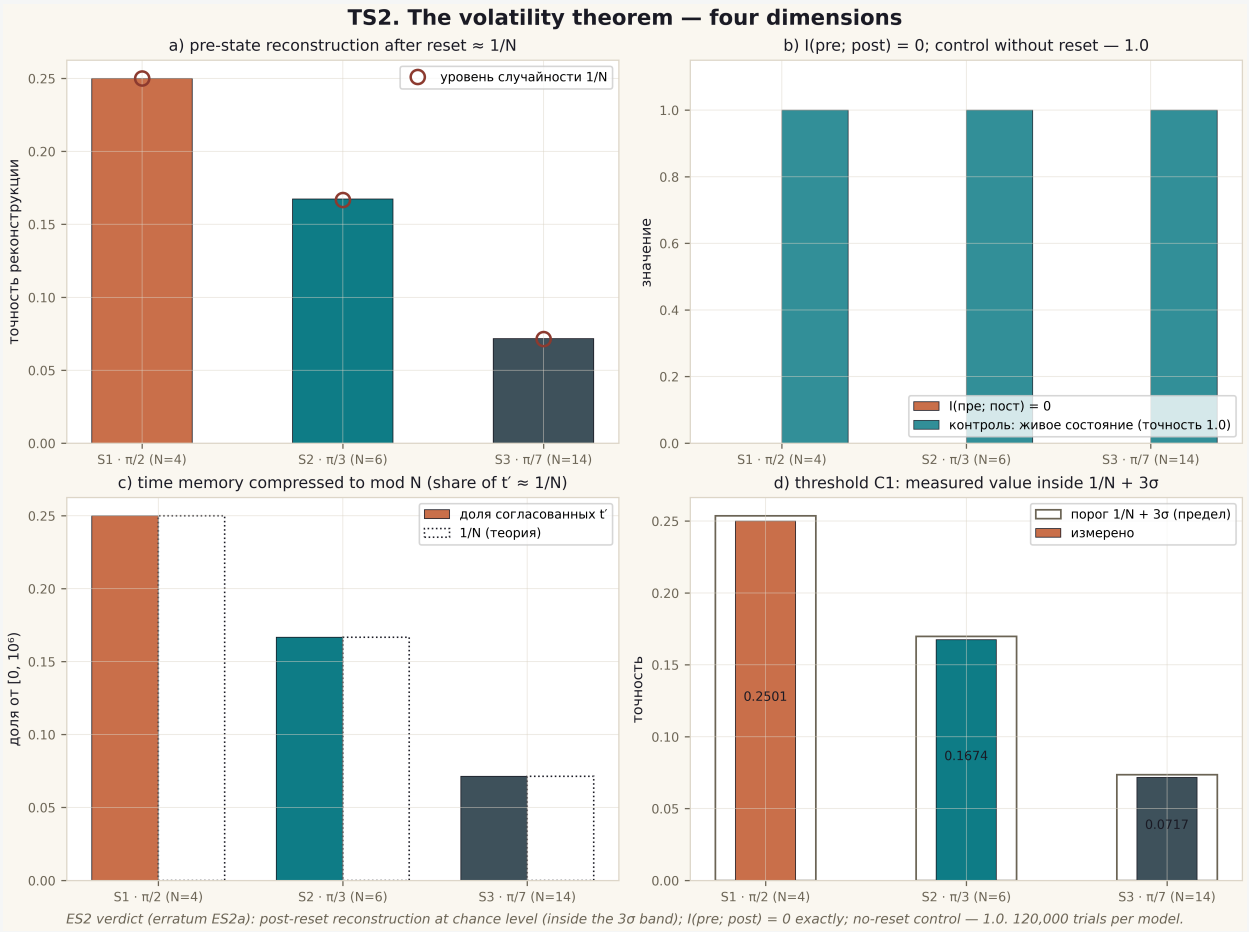


Fig. 1. Theorem TS2: four dimensions — post-reset reconstruction, mutual information, time compression to mod N, the 3σ threshold (2×2 tile, 600 dpi, ES2 data).

Boundaries and honest limitations

The theorem certifies information-theoretic irreversibility of erasing the volatile register in the model. It says nothing about the thermodynamic cost of erasure (Landauer's principle), about physical carriers, or about readout noise — the channel robustness boundary is measured separately (ES5). Erratum ES2a is recorded in the verdict: the primary C1 threshold was replaced by the 3σ band before the verdict was accepted.